

AT2: Predictive Modelling and Portfolio Evaluation 50 Points Possible

09/10/2026

Attempt 1 

In Progress

NEXT UP: Submit Assignment

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Unlimited Attempts Allowed

17/08/2026

▼ Details



Assessment Details

- **Due:** Week 7
- **Weighting:** 50%
- **Format:** The submission consists of the following: R script; a recorded video presentation explaining the analysis and recommendations to management and other stakeholders; executive summary
- **Type/Product:** Written and oral piece
- **Length/Duration:** A 250-word written reflection and a recorded video presentation (6–10 minutes).
- **Individual/Group:** Individual

Learning Outcomes Assessed:

- **CLO1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO2:** Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.
- **CLO3:** Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.
- **CLO4:** Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

Task Description

- **Goal:** To apply quantitative financial analytics techniques in R to evaluate portfolio risk and return, assess long-term investment viability, and communicate data-driven insights to both technical and non-technical stakeholders.
- **Description:** You will select two large-cap stocks of your choice and retrieve historical monthly price data using R. You are required to estimate beta, apply the Capital Asset Pricing Model (CAPM), assess portfolio diversification through return correlation, produce a six-month price forecast using a method of your choice, and discuss how derivative securities can be used to manage downside risk. You will also evaluate a hypothetical

five-year investment project by computing its Net Present Value (NPV) and Internal Rate of Return (IRR), and extend your analysis to account for uncertainty in projected cash flows.

□ AI-Supported Learning Statement

This task permits the responsible and ethical use of artificial intelligence (AI) tools to support your learning and completion of some aspects of this task.

The work you submit for this task must remain your own.

This means:

- The ideas, arguments, and final responses you submit must be your own work.
- Your submission must reflect your understanding, voice and learning.
- You are responsible for checking the accuracy, relevance, and ethics of any AI-assisted content you use.
- You must acknowledge the use of AI in your submission.

Please refer to the [Artificial Intelligence for Students Guide \(https://libguides.usc.edu.au/ai-students\)](https://libguides.usc.edu.au/ai-students) for guidance on how to use AI responsibly and how to [acknowledge your use of AI \(https://libguides.usc.edu.au/ai-studentguide/responsible#s-lg-box-23112449\)](https://libguides.usc.edu.au/ai-studentguide/responsible#s-lg-box-23112449).

While you may use AI as outlined in the assessment instructions, you must ensure that the authorship of the work remains your own. Do not hand over (transfer) or give up (relinquish) authorship, and do not falsify or fabricate any content or supporting documents.

Assessment Support and Resources

The following resources are provided to support you in this assessment task:

- Task Description Sheet: [Assessment2.pdf \(https://learn.usc.edu.au/courses/35359/files/3075856?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3075856?wrap=1)

▼ View Rubric

Assessment 2

Criteria	Ratings						Pts
Financial Analysis — Question 1 view longer description	20 to >17 pts High Distinction (85% or above) Beta is correctly estimated and CAPM is applied accurately for both stocks. The SML is used insightfully to assess over- and undervaluation with clear interpretation of what this means for an investor. The correlation	17 to >15 pts Distinction (75% to 84.99%) Beta, CAPM, and SML are correctly applied with clear interpretation. Correlation is computed and diversification implications are discussed with minor gaps in depth. A forecasting method is applied and justified, with reasonable evaluation of reliability. The	15 to >13 pts Credit (65% to 74.99%) Beta and CAPM are computed correctly and the SML is used to classify stocks, but interpretation lacks depth. Correlation is computed but its portfolio implications are not fully developed. A forecast is produced but the justification for the chosen	13 to >10 pts Pass (50% to 64.99%) Beta and CAPM are computed, possibly with minor errors. The SML is referenced but not meaningfully applied. Correlation may be computed without interpretation. A forecast is produced but method choice is not justified and reliability	10 to >0 pts Fail (below 50%) One or more core components of Question 1 are missing or incorrectly executed. Analysis is superficial or demonstrates fundamental misunderstanding of the concepts involved. Little or no connection is made between findings and	0 pts No Marks	/ 20 pts

Assessment 2

Criteria	Ratings					Pts	
	between stock hedging returns is computed and accurate and its implications covers both for portfolio diversification are discussed with depth. A well-justified forecasting method is applied, the forecast is plotted clearly, and its reliability is critically evaluated. The hedging discussion identifies an appropriate derivative instrument, explains its mechanics clearly, and honestly reflects on its practical limitations. All findings are linked back to a concrete recommendation for the portfolio manager.	method is limited and reliability is not critically assessed. The hedging discussion is accurate but surface-level. Findings are summarised but the link to investment decisions is weak.	is not discussed. The hedging discussion is minimal. Findings are reported but not connected to a clear recommendation.	investment decisions.			
Financial Analysis — Question 2 view longer description	10 to >8.5 pts High Distinction (85% or above) NPV and IRR are correctly computed and interpreted clearly in plain language. The student engages substantively with the question of cash flow uncertainty, either by providing a well-justified argument for	8.5 to >7.5 pts Distinction (75% to 84.99%) NPV and IRR are correctly computed and interpreted. The issue of cash flow uncertainty is acknowledged and addressed with reasonable depth. The approach taken is justified with only minor gaps. Findings are visualised to	7.5 to >6.5 pts Credit (65% to 74.99%) NPV and IRR are correctly computed but interpretation is limited. Cash flow uncertainty is acknowledged but not meaningfully addressed beyond the base case. A visualisation is included but may be basic. The recommendation is not clearly	6.5 to >5 pts Pass (50% to 64.99%) NPV and IRR are computed, possibly with minor errors. Little or no engagement with the question of cash flow uncertainty. Visualisation may be absent or uninformative. The recommendation is minimal or	5 to >0 pts Fail (below 50%) NPV and IRR are missing or incorrectly computed. No meaningful engagement with uncertainty. No recommendation is provided or it is wholly disconnected from the analysis.	0 pts No Marks	/ 10 pts

Assessment 2

Criteria	Ratings						Pts
	why the base case is sufficient, or by extending the analysis to reflect uncertainty and providing management with a fuller picture of possible outcomes. The chosen approach is explicitly justified. Findings are presented visually and the written recommendation to management is clear, evidence-based, and accessible to a non-technical audience.						
R Code Quality and Reproducibility view longer description	10 to >8.5 pts High Distinction (85% or above) Code covering both questions is fully reproducible, logically structured, and uses meaningful variable names. Clear comments explain what each key step does and why. Appropriate R functions and packages are used throughout. No unnecessary or redundant code is present. All outputs directly	8.5 to >7.5 pts Distinction (75% to 84.99%) Code is reproducible and sensibly structured with mostly appropriate commenting. Variable naming is generally clear. Minor inefficiencies or small clarity issues may exist but the code remains fit for purpose and supports the analysis across both questions.	7.5 to >6.5 pts Credit (65% to 74.99%) Code runs and produces correct outputs for both questions but lacks clarity or consistent commenting. Variable names may be unclear and some redundant steps may be present. The analytical intent is understandable but the code would be difficult to reuse or adapt.	6.5 to >5 pts Pass (50% to 64.99%) Code runs with some difficulty and is poorly structured. Comments are minimal or absent. Outputs may be partially correct but are not cleanly aligned with the analysis or visualisations presented. Coverage of one question may be substantially weaker than the other.	5 to >0 pts Fail (below 50%) Code is missing, incomplete, or not reproducible for one or both questions. It does not run, produces incorrect outputs, or is so poorly structured that it cannot support the analysis.	0 pts No Marks	/ 10 pts

Assessment 2

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	support the analysis and findings presented in the video and executive summary.						
Video Presentation and Executive Summary view longer description	10 to >8.5 pts High Distinction (85% or above) Video: Student is visible on camera throughout and shows their student ID at the start. The presentation is clear, professional, and engaging, running between 6 and 10 minutes. It covers the approach taken for both questions, explains why each analytical choice was made, connects the code to the conclusions, and presents key findings and recommendations in a way that is accessible to a non-technical management audience. Executive summary: Is approximately 250 words, well written, and clearly addresses all four required questions (what was analysed, what was found, what is	8.5 to >7.5 pts Distinction (75% to 84.99%) Video: Covers clearly, explains the approach and connects code depth, or to conclusions, with only minor gaps in depth or delivery. All presentation requirements are met. Executive summary: Addresses all four required questions with only minor weaknesses in clarity or depth. Word count is approximately 250 words.	7.5 to >6.5 pts Credit (65% to 74.99%) Video: Covers the required content for both questions but lacks structure, depth, or professional delivery. One or more presentation requirements (e.g., student ID, duration) may not be fully met. Executive summary: Addresses most required questions but is uneven in coverage or depth.	6.5 to >5 pts Pass (50% to 64.99%) Video: Is difficult to follow, does not clearly explain the approach or connect findings to conclusions, or is significantly outside the required duration. Coverage of one question may be substantially weaker than the other. Executive summary: Only partially addresses the required questions.	5 to >0 pts Fail (below 50%) Video: Is missing or wholly inadequate. Required content is not addressed or the presentation demonstrates no meaningful engagement with the task. Executive summary: Is missing or fails to engage meaningfully with the required questions.	0 pts No Marks	/ 10 pts

Assessment 2

Criteria	Ratings	Pts
	recommended, and what the limitations are) drawing on findings from both questions.	
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